Resolution of ALARM_5013 Route Flap Issue on Huawei Access Node YEL-ACC-22

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5013

Description

The ALARM_5013 Route Flap issue on the Huawei Access Node YEL-ACC-22 at Yelahanka Sub-station is a critical failure domain that necessitates immediate attention and resolution. This failure domain is primarily rooted in the Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) phenomenon, which disrupts the stable operation of the routing protocols and subsequently impacts the overall network connectivity and availability. The underlying protocol or hardware mechanics responsible for this failure domain are the dynamic routing protocols employed by the Huawei Access Node, including BGP, OSPF, and ISIS, which are prone to flapping due to various factors such as misconfiguration, link instability, or hardware issues. Furthermore, the Huawei Access Node's architecture and design, including its routing table management and protocol interaction mechanisms, also contribute to the likelihood of this failure domain. Therefore, it is essential to identify and rectify the root causes of this issue to prevent future occurrences and ensure seamless network operation.

Pre-requisite Checks

- Check: Verify the Node ID YEL-ACC-22 is accessible via the Huawei CLI

- Check: Confirm the Site Location Yelahanka Sub-station is properly configured and connected

- Check: Ensure the Huawei Access Node is powered on and functioning within normal operating parameters

- Check: Verify the redundant link status on the Huawei Access Node

Potential Root Causes

Cause	Probability
Misconfigured BGP peering on the Huawei Access Node	High
Link instability between the Huawei Access Node and the neighboring node	Medium
Hardware failure on the Huawei Access Node	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: Peer State: Established, Description: This command displays the BGP peer status, which helps identify any issues with the BGP peering relationship between the Huawei Access Node and its neighboring nodes. The expected output of this command is an 'Established' peer state, indicating a stable BGP connection.
```

```
Command: display ospf neighbor, Expected Output: Neighbor State: Full, Description: This command displays the OSPF neighbor status, which helps identify any issues with the OSPF adjacency between the Huawei Access Node and its neighboring nodes. The expected output of this command is a 'Full' neighbor state, indicating a stable OSPF connection.

Command: display isis lsp, Expected Output: LSP State: Up, Description: This command displays the ISIS LSP (Link-State Protocol) status, which helps identify any issues with the ISIS protocol operation on the Huawei Access Node. The expected output of this command is an 'Up' LSP state, indicating a stable ISIS connection.
```

Resolution Steps

Step 1: Step: Verify and Correct BGP Peering Configuration, Command: undo bgp peer timer refresh, Description: This command corrects the BGP peering configuration by resetting the BGP peer timer, which helps resolve issues related to BGP peering instability. The expected output of this command is an 'OK' confirmation message, indicating a successful configuration update.

Step 2: Step: Verify and Correct OSPF Neighbor Configuration, Command: undo ospf neighbor timer refresh, Description: This command corrects the OSPF neighbor configuration by resetting the OSPF neighbor timer, which helps resolve issues related to OSPF adjacency instability. The expected output of this command is an 'OK' confirmation message, indicating a successful configuration update.

Step 3: Step: Verify and Correct ISIS LSP Configuration, Command: undo isis lsp timer refresh, Description: This command corrects the ISIS LSP configuration by resetting the ISIS LSP timer, which helps resolve issues related to ISIS protocol instability. The expected output of this command is an 'OK' confirmation message, indicating a successful configuration update.

Post-Check Verification

```
Command: display bgp peer, Expected Output: Peer State: Established, Description: This command verifies the BGP peer status after the resolution steps have been completed. The expected output of this command is an 'Established' peer state, indicating a stable BGP connection.

Command: display ospf neighbor, Expected Output: Neighbor State: Full, Description: This command verifies the OSPF neighbor status after the resolution steps have been completed. The expected output of this command is a 'Full' neighbor state, indicating a stable OSPF connection.

Command: display isis lsp, Expected Output: LSP State: Up, Description: This command verifies the ISIS LSP status after the resolution steps have been completed. The expected output of this command is an 'Up' LSP state, indicating a stable ISIS connection.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: rollback to , Description: This command restores the original configuration to its previous state, which helps roll back the changes made during the resolution steps. The expected output of this command is an 'OK' confirmation message, indicating a successful rollback.